NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 29/03/2026
Time: 3 Hrs | GTM – 08 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	B	A	C	C	D	D	D	A	C
11	12	13	14	15	16	17	18	19	20
C	C	C	B	D	B	C	C	A	D
21	22	23	24	25					
0	3	9	8	2					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	C	B	A	B	C	A	A	D	A
36	37	38	39	40	41	42	43	44	45
A	C	C	A	B	B	C	A	B	C
46	47	48	49	50					
3	3	8	75	45					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	C	D	D	A	C	D	B	D	A
61	62	63	64	65	66	67	68	69	70
A	D	B	C	C	D	A	A	D	A
71	72	73	74	75					
23	33	10	109	244					

Rankers Academy JEE

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HINTS
PHYSICS

1. $2g - T = 2a$

$T - g = a$

$\Rightarrow 2g - T = 2(T - g)$

$3T = 4g$

$T = \frac{4g}{3}$

$1000 \times x = \frac{4 \times 10}{3}$

$\Rightarrow x = \frac{4}{3} cm$

2. $\frac{m(\sqrt{3}) + (n-1)mT\sqrt{3}}{nm} = \frac{u^2 \sin 2\theta}{g}$

$\Rightarrow \frac{\sqrt{2} + (n-1)7\sqrt{3}}{n} = \frac{100}{10} \times \frac{\sqrt{3}}{2}$

$\Rightarrow 1 + (n-1)7 = 5n$

$1 + 7n - 7 = 5n$

$2n = 6$

$n = 3$

3. Ratio $= \frac{\sqrt{(5l)^2 + (10l)^2}}{5l} = \sqrt{5}$.

4. $x = \sqrt{2gy}\sqrt{2\frac{(2h - y)}{g}}$

$\Rightarrow x = \sqrt{4(2hy - y^2)}$

X is maximum if $\frac{d}{dy}(2hy - y^2) = 0$

i.e y=h.

5. Change of state takes place at 20°C.

$\Rightarrow mL = pt$

$2L = (5kJ)2$

$\Rightarrow L = 5kJ / kg$

6. $l = 2\lambda$

$\lambda = 4d$

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$$\Rightarrow \frac{l}{d} = 8$$

7. Theoretical concept.

8. $\overline{(A+B).C}$

9.

$$q = q_o \cos wt$$

$$i = \frac{-dq}{dt} = q_o \omega \sin wt$$

$$\frac{di}{dt} = q_o \omega^2 \cos wt$$

$$\left(\frac{di}{dt}\right)_{max} = q_o \omega^2$$

10. D = 96 cm

If lens displacement is 'L'

$$\Rightarrow m_1 = \frac{D+L}{D-L}, m_2 = \frac{D-L}{D+L}$$

$$\Rightarrow \left(\frac{D+L}{D-L}\right)^2 = 4.84$$

$$\frac{D+L}{D-L} = 2.2 = \frac{11}{5}$$

$$5D + 5L = 11D - 11L$$

$$6D = 16L$$

$$3D = 8L$$

$$\Rightarrow f = \frac{D^2 - L^2}{4D}$$

$$= \frac{D^2 - L^2}{4D}$$

$$= \frac{55D}{64 \times 4} = \frac{55 \times 96}{64 \times 4} = 20.6cm$$

11. $$\text{Least count} = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

$$= \frac{1\text{mm}}{100} = 0.01\text{mm} = 0.001\text{cm}$$

Zero error = -0.33 mm

Measurement = 3.76-(-0.03)=3.79 mm

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12. **(Reference NCERT 11.23)**

$$T = \frac{PV}{nR} = \frac{P\sqrt{V}\sqrt{V}}{nR} = \frac{K\sqrt{V}}{nR}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{V_1}{V_2}} = \frac{1}{\sqrt{2}}$$

13. **(Reference NCERT 7.33)**

$$F = \frac{GMmh}{\left(r^2 + h^2\right)^{\frac{3}{2}}}$$

$$F^{|} = \frac{GMm.2h}{\left(r^2 + 4h^2\right)^{3/2}}$$

$$h = r$$

$$\frac{F^{|}}{F} = \frac{4\sqrt{2}}{5\sqrt{5}}$$

14. $2mgl = \frac{1}{2}\frac{2m}{3}\left(V_1 + V_2\right)^2 \Rightarrow \left(V_1 + V_2\right) = \sqrt{6gl}$.....(1)

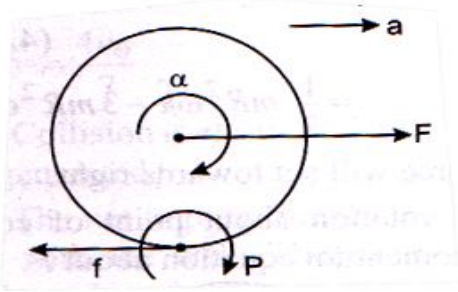
$mV_1 = 2mV_2 \Rightarrow V_1 = 2V_2$.....(2)

From (1) & (2)

$$V_1 = = \sqrt{\frac{8gl}{3}}$$

15. As sphere is rolling we can apply torque equation about point of constant (about P)

$$FR = I_p\alpha$$



$$\Rightarrow FR = \left(\frac{7}{5}MR^2\right)\alpha$$

$$\Rightarrow R\alpha = \frac{5F}{7M} = a \text{ (acceleration of CM)}$$

Equation of translator motion

$$F - f = mA \Rightarrow f = \frac{2}{7}F \qquad \text{.(i)}$$

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For no sliding, $f \le \mu mg \Rightarrow F = \frac{7}{2}\mu mg$

16. $\Delta x = (\mu_A - 1)t_A - (\mu_B - 1)t_B$

$\Delta x = t_B - t_A$

17. As, $a_2 > a_1$ hence $v_1 > v_2$ As the liquid is non-viscous, there is no loss of energy, i.e., energy per unit volume of the liquid is the same at every point. So option (3) is incorrect.

18. $\oint E_p dS = \frac{Q+Q'}{\varepsilon_0}$ where Q' is the charge outside the sphere

$$Q_1' = \int_R^r dQ = \int_R^r \frac{\alpha}{r} \times 4\pi r^2 dr$$

$$E_p \times 4\pi r^2 = \frac{Q + 2\pi\alpha\left(r^2 - R^2\right)}{2r^2\varepsilon_0}$$

E is independent of r if $\frac{Q}{4\pi r^2\varepsilon_0} - \frac{\alpha R^2}{2r^2\varepsilon_0} = 0$

$Q = 2\pi R^2\alpha$

19. Due to electrostatic shielding, the ratio will be one

20 The equivalent resistance $= \left[7 + \frac{3R}{3+R}\right]$

The power consumed $\frac{24^2}{\left[7 + \frac{3R}{3+R}\right]} = 60 \Rightarrow R = 19.5\Omega$

21. Use kirchoff's laws. & symmetry

22. $i = \frac{E}{\sqrt{R^2 + X_C^2}}$

$$\frac{i}{2} = \frac{E}{\sqrt{R^2 + 9X_C^2}}$$

$$2 = \frac{\sqrt{R^2 + 9X_C^2}}{\sqrt{R^2 + X_C^2}}$$

$$\Rightarrow 4R^2 + 4X_C^2 = R^2 + 9X_C^2$$

$$3R^2 = 5X_C^2$$

$$\Rightarrow \frac{X_C}{R} = \sqrt{\frac{3}{5}}$$

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23. $P = \frac{F}{A} = \left[2\times(0.2)+0.8\right]\frac{I}{C}.$

$$F = \frac{(1.2)P}{C} = \frac{1.2\times12}{3\times10^8} = 4.8\times10^{-8}$$

$\Rightarrow \lambda = 1, n = 8$

$\Rightarrow \lambda + n = 9$

24. If the current in the solenoid is i, the magnetic field inside the solenoid is $B = \mu_0 ni$ parallel to its axis. Outside the solenoid, the field will be zero. The flux of the magnetic field through the coil will be $\Phi = B\pi r^2 N$ where r is the radius of the solenoid and N is the number of turns in the coil. The induced emf will have magnitude

$$\frac{d\Phi}{dt} = N\ \pi r^2 \frac{dB}{dt} = \pi r^2 N\mu_0 n\frac{di}{dt}$$

If R denotes the resistance of the coil, the current through the galvanometer is

$$I = \frac{\pi r^2 N}{R}\mu_0\, n\frac{di}{dt}$$

Or $Idt = \frac{\pi r^2 N}{R}\mu_0 n\ di.$

The total charge passing through the galvanometer is

$$\Delta Q = \int Idt = \frac{\pi r^2 N}{R}\mu_0 n\int di$$

$$= \frac{\pi r^2 N\mu_0 n}{R}\Delta i$$

$$= \frac{\pi(2cm)^2\times100\times4\pi10^{-7}\frac{T-m}{A}\times100cm^{-1}\times10A}{20\Omega}$$

$$= 8\times10^{-4}C = 800\mu C$$

25. $v = v_1 + v_2 + v_3 + v_4 + ...$

$$= \frac{GM}{1}\ \frac{GM}{2}\ \frac{GM}{4}\ \frac{GM}{8}\ \frac{GM}{16}\ \frac{GM}{32}\ .....$$

$$= GM\left(1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+.....\right)\left(1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+.....\right)[0,1]$$ □ G.P. of infinite series

$$= 1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+....$$

$$= \frac{1}{1-\frac{1}{2}} = 2$$

$V = -GM(2)$ $= -2GM$

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CHEMISTRY

26. Equivalent weight = M.W/Valency factor

26. Conceptual

27. Possible values of l and m depend upon the value of n

$l = 0\ to\ (n-1)$

$m = -1 to +l$ through zero

$s = +\frac{1}{2}\ and\ -\frac{1}{2}$

Thus for n = 3,

l may be 0, 1 or 2; but not 3

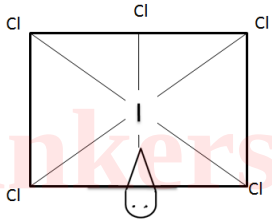
m may be -2, -1, 0, +1 or +2

s may be $+\frac{1}{2}\ or\ -\frac{1}{2}$

28.

For $^{**}I\,Cl_5$

$H = \frac{1}{2}(7+5-0+0) = 6(sp^3d^2)$



sp^3d^2 hybridized

Geometry : Octahedral

Shape / Structure : Square pyramidal

For $^{**}I\,Cl_4^-$

$H = \frac{1}{2}(7+4-0+1) = 6(sp^3d^2)$

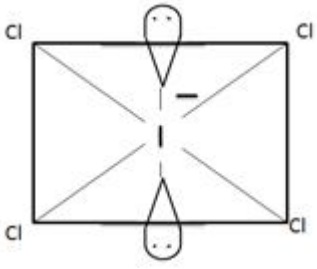
sp^3d^2 hybridized

Geometry : Octahedral

Shape / Structure : Square planar

So, ICl_5 and ICl_4^- are isolobal but not isostructural.

29.



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Species	Mo energy order	Bond order	n, number of unpaired e^-	Magnetic character
$C_2^{2-}\left(14e^-\right)$	$\left[8e^-\right]\pi_{2P_x^2}=\pi_{2P_y^2}\sigma_{2P_z^2}$	$\frac{6-0}{2}=3$	0	Diamagnetic
$O_2\left(16e^-\right)$	$\left[8e^-\right]\sigma_{2P_z^2}\ \pi_{\pi_{2P_x^2}}=\pi_{2P_y^2}\ \pi^*_{\pi_{2P_x^1}}=\pi^*_{2P_y^1}$	$\frac{6-2}{2}=2$	2	Paramagnetic
$O_2^{2-}\left(18e^-\right)$	$\left[8e^-\right]\sigma_{2P_z^2}\ \pi_{2P_x^2}=\pi_{2P_y^2}\ \pi^*_{2P_x^2}=\pi^*_{2P_y^2}$	$\frac{6-4}{2}=2$	0	Diamagnetic
$N_2^{2-}\left(16e^-\right)$	$\left[8e^-\right]\pi_{2P_x^2}=\pi_{2P_y^2}\sigma_{2P_z^2}\ \pi^*_{2P_x^1}=\pi^*_{2P_y^1}$	$\frac{6-2}{2}=2$	2	Paramagnetic

Bond length $=\alpha\frac{1}{BO(Bond\ order)}$

So order of bond length $\underset{(BO=3)}{C_2^2}<\underset{(BO=2)}{O_2}=N_2^{2-}<\underset{(BO=1)}{O_2^{2-}}$

The diamagnetic species with shortest bond length is C_2^{2-} (option-a)

30. 3,4,5

31. Moles of urea = $\frac{5}{60}$;

Moles of fructose = $\frac{5}{180}$;

$C_{urea}=\frac{0.08\times1000}{100}\Rightarrow0.833$

Moles of sucrose = $\frac{5}{342}$;

$C_{sucrose}=0.0146\times\frac{1000}{100}=01.46$

Moles of KCl(effective)

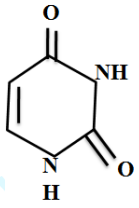
$=2\times\frac{5}{74.5}=\frac{5}{37.25};\frac{0.134\times1000}{100}=1.34$

$\pi\alpha C$

This order is $\pi_4>\pi_1>\pi_2>\pi_3$

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32. (RNA) contains, adenine (A), guanine (G), cytosine (C) and uracil (U). In the given options



Is uracil (present in RNA only)

33.

$$\eta = \frac{T_2 - T_1}{T_2} = 1 - \frac{T_1}{T_2}$$

$$= 1 - \frac{373}{673} = 0.446$$

$$\eta = \frac{\Delta w}{q_1}$$

or $\Delta W = \eta \times q_1 = 0.446 \times 3.347 = 1.49$ kJ

34. $PbCl_2$ is insoluble in cold water, soluble in hot water and PbS is black ppt in acidic medium.

35. Conceptual

36. (a) The spin only magnetic moment (μ) of each ion can be calculated as

$$\mu = \sqrt{n(n+2)}\,BM$$

$[\because n = No.of\ unpared\ electron(s)] \Rightarrow \mu\ \alpha$ n, i.e,

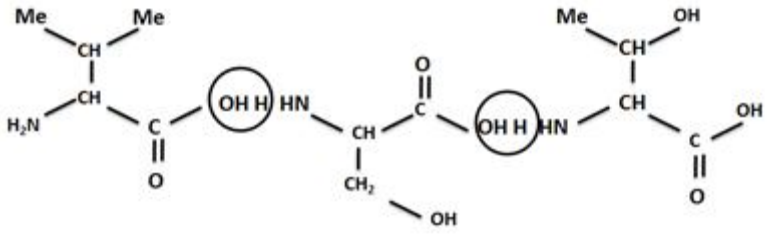
Higher the number of unpaired electrons, higher will be the value of μ

Metal ion	Z	n (for metal ion)	M (BM)	Nature
Ti^{2+}	22	$2(3d^2)$	$\sqrt{8}$	Paramagnetic
V^{2+}	23	$3(3d^3)$	$\sqrt{15}$	Paramagnetic
Ti^{3+}	22	$1(3d^1)$	$\sqrt{3}$	Paramagnetic
Sc^{3+}	21	$0(3d^0)$	2	Diamagnetic

Thus, the correct order of spin only magnetic moments of given hydrated ions will be

$Sc^{3+} < Ti^{3+} < Ti^{2+}\ V^{2+}$

37. Formation of the tripe tide (Val-Ser-Thr) can be shown as:



38. In homoleptic complexes, the metal atom/ion is linked to only one type of ligand. Assuming, ligands are neutral, the octahedral complexes of M^{3+} can be, $[M(L_1)_6]^{3+}$, $[M(L_2)_6]^{3+}$ and $[M(L_3)_6]^{3+}$

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(I) (II) (III)
Green Blue Red (wave length)

$\lambda_{Absorption}$

So, $\lambda_{III}^{L3} > \lambda_{I}^{L1} > \lambda_{II}^{L2}$

$\therefore \Delta_{Absorption}^{o} : \Delta_{II}^{L2} > \Delta_{I}^{L1} > \Delta_{III}^{L3}$

$\left[\because Enegry\left(\Delta, CFSE\ \alpha \frac{1}{\lambda}\right)\right]$

We know, ligand strength $\alpha\ \Delta_{Absorption}^{0}$

So, the increasing order of the ligand strength will be, $L_3 < L_1 < L_2$

39. $y_A = \dfrac{P_A^0 X_A}{P_A^0 X_A + P_B^0 X_B}$

$y_A = \dfrac{1}{1 + \dfrac{P_B^0 X_B}{P_A^0 X_A}}$

$y_A = \dfrac{1}{1+(3)(3)} = \dfrac{1}{10} = 0.1$

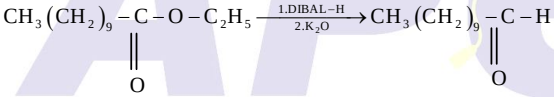
40. More positive E_{SRP}^{0} :Good OA

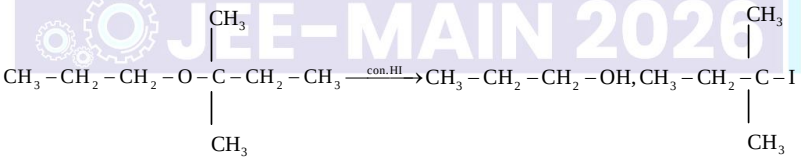
41. Methanamine $\underset{3.38}{Pkb}$ Ethanamine 3.29

N_1N -diethylethanamine 9.38

N – ethyl ethanamine 3.00

42. The IUPAC name of allybromide is 3-Bromo-propene and tert- Butylbromide is 2-Bromo-2-methyl propane

43.


44.


45. Conceptual

46. $\left[Ni\left(NH_3\right)_6\right]^{2+}$ has sp^3d^2 hybridization and contains two unpaired electrons.

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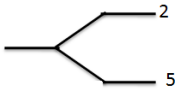
$\mu = \sqrt{2(2+2)} = \sqrt{2\times 4} = \sqrt{8} = 2.83\,B.M$

$\cong 28\times 10^{-1}\,B.M$

47. Odd electron species are para magnetic

48. Calculation of C-H BDE from CH_4 and apply in C_2H_6

49. NCERT – XI, VOL – II, P.NO: 358



$\omega\%$ of

$N = \dfrac{1.4\times 10\times 2}{0.5} = 56\%$

$56 = \dfrac{14\times 3}{G.M.W(ing)}\times 100 \Rightarrow G.M.W = 75g$

50. Conceptual

MATHEMATICS

51. $y(\alpha) = \sec\alpha + \tan\alpha$

$\dfrac{dy}{d\alpha} = \sec\alpha\ \tan\alpha + \sec^2\alpha$

$\left(\dfrac{dy}{d\alpha}\right)_{\alpha=\frac{\pi}{6}} = 2$

52. $D>0\quad (a+1)^2 - 4(a^2+a-8) > 0$

$a^2 + 2a + 1 - 4a^2 - 4a + 32 > 0$

$-3a^2 - 2a + 33 > 0$

$3a^2 + 2a - 33 < 0$

$3a^2 + 11a - 9a - 33 < 0$

$a(3a+11) - 3(3a+11) < 0$

$\dfrac{-11}{3} < a < 3$

And $a.f(2) < 0$

$1\times(4 - 2(a+1) + a^2 + a - 8) < 0$

$4 - 2a - 2 + a^2 + a - 8 < 0$

$a^2 - a - 6 < 0$

$a^2 - 3a + 2a - 6 < 0$

$(a-3)(a+2) < 0$

$-2 < a < 3$

53. equation of tangent at θ is

$\dfrac{x\sec\theta}{2} - \dfrac{y\tan\theta}{\sqrt{2}} = 1$

$P = (2\cos\theta, 0), Q = (0, -\sqrt{2}\cot\theta)$

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$h=2\cos\theta \qquad k=-\sqrt{2}\cot\theta$

Eliminating θ we get $\frac{4}{x^2}-\frac{2}{y^2}=1$

54. $IF=1+x^2, y\left(1+x^2\right)=\frac{4x^3}{3}+c$

55. $n(E)=53\times 4!, n(S)=7!-6!=6(6!)$

$P(E)=\frac{53\times(4!)}{6(6!)}=\frac{53}{180}$

56.

57.

(3)	(4)	(4)	(3)
A	B	C	D
0	3	3	0
1	2	2	1
2	1	1	2
3	0	0	3

Required number of ways = 485

58.

$\frac{\left(\bar{b}.\bar{a}\right)\bar{a}}{|a|^2}=\frac{4}{3}\left(\bar{i}-\bar{j}-\bar{k}\right)$

$(\lambda+2)\left(\bar{i}-\bar{j}-\bar{k}\right)=4\left(\bar{i}-\bar{j}-\bar{k}\right)$

$\lambda=2$

59. $|B|=\begin{pmatrix} a_{11} & 4a_{21} & 16a_{31} \\ 4a_{12} & 16a_{22} & 64a_{32} \\ 16a_{13} & 64a_{23} & 256a_{33} \end{pmatrix} =4^6\begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix}=4^6|A|$

$|A|=\frac{256}{4^6}=\frac{1}{16}$

60. $R=\left(1,\frac{1+3\lambda}{1+\lambda}\right)$. It is an interior point of $y^2-4x=0$ if $\left(\frac{1+3\lambda}{1+\lambda}\right)^2$ -4<0

Therefore, $-\frac{3}{5}<\lambda<1$. But $\lambda>0$

61. Let $P=\left(\sqrt{2}a\cos\phi,\sqrt{2}b\sin\phi\right).F_1 \text{ and } F_2=\left(\pm\sqrt{2}ae,0\right)$

The area of

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$$\Delta PFF' = \left|\frac{1}{2}\begin{vmatrix} \sqrt{2}a\cos\phi & \sqrt{2}b\sin\phi & 1 \\ \sqrt{2}ae & 0 & 1 \\ -\sqrt{2}ae & 0 & 1 \end{vmatrix}\right| = \frac{1}{2}.\sqrt{2}b\sin\phi.\sqrt{2}ae = 2abe\sin\phi$$

$\therefore$ maximum area=2ab. $\frac{\sqrt{a^2-b^2}}{a}$.

62. Conceptual

63. $\therefore g\{f(x)\} = \{f(x)\}^2 + 1, f(x) \neq 2$

$3, f(x) = 2$

$\therefore g\{f(x) = \sin^2 x + 1, x \neq n\pi$

$3, x = n\pi$

RH limit = $\lim\limits_{x\to 0^+} g\{f(x)\} = \lim\limits_{h\to 0} g\{f(0+h)\} = \lim\limits_{h\to 0}\{\sin^2(0+h)+1\} = 1$

LH = limit $\lim\limits_{x\to 0+0} g\{f(x)\} = \lim\limits_{h\to 0} g\{f(0+h)\} = \lim\limits_{h\to 0}\{\sin^2(0-h)+1\} = 1$

64. Conceptual

65. $a^2 - 4a + 6 = (a-2)^2 + 2 > 1$ for all a. So, $\min\limits_{a\in R}(1, a^2 - 4a + 6) = 1$

$\therefore$ the equation is $\sin x + \cos x = 1$

$$\sqrt{2}\left[\frac{1}{\sqrt{2}}.\sin x + \frac{1}{\sqrt{2}}.\cos x\right] = 1$$

$$\sin\left[x + \frac{\pi}{4}\right] = \sin\frac{5}{4}$$

$$x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$$

66. Put $x = i$

Then $(a_0 - a_2 + a_4 - a_6 + ...) + i(a_1 - a_3 + a_5 - a_7 + ...) = 0$

$(1)(2)$ are true (by comparison of real parts and Im. Parts, both (1) (2) are true.

$(\because a_0 = 1)$

Put $x = -1$, then $a_0 - a_1 + a_3 - a_3 + ... + a_{30} = 0$

$\therefore (3)$ is false

67. $a + 10 = (a+20)2^5 \Rightarrow -31a = 630 \Rightarrow$ a = -630/31

Now, $a + 20 = \frac{-630}{31} + 20 = -\frac{10}{31}$

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68. $\therefore$ Equation of chord of contact is $(4y-1)+\alpha(x-2y)=0$

69. $\int$ *Let* $x=t^6$

$$\int\frac{\left(t^6+t^4+t\right)6t^5dt}{t^6\left(t^2+1\right)}=6\int\frac{t^5+t^3+1}{t^2+1}$$

$$6\int\left(t^3+\frac{1}{t^2+1}\right)=\frac{3t^4}{2}+6\tan^{-1}t$$

70. Using Leibnitz rule

71. $$A=\begin{pmatrix}1&2&3\\1&3&5\\2&5&a\end{pmatrix}\quad B=\begin{pmatrix}6\\9\\b\end{pmatrix}$$

$(adjA)B=0$

72.

73. Given, $n(A) = m$ and $n(B) = n$

$\therefore$ Number of subsets of $A = 2^m$

And number of subsets of $B = 2^n$

According to the question,

$2^m - 2^n = 56$

$\Rightarrow 2^m - 2^n = 2^6 - 2^3$

$\Rightarrow \quad m=6$ and $n=3$

So, distance between the points P(m, n), i.e. P (6, 3) and Q(-2, -3) is given by

$$\sqrt{(6+2)^2+(3+3)^2}=\sqrt{64+36}$$
$$=\sqrt{100}=10$$

74. $\sin x=-\frac{3}{5}$ and $\pi<x<\frac{3\pi}{2}$

So, $\cos x=-\frac{4}{5}$ and $\tan x=\frac{3}{4}$

Then,

$$80(\tan^2 x-\cos x)=80\left[\left(\frac{3}{4}\right)^2-\left(-\frac{4}{5}\right)\right]$$

$$=80\left[\frac{9}{16}+\frac{4}{5}\right]=80\left[\frac{45+64}{80}\right]\qquad=80\times\frac{109}{80}=109$$

75. $z^2+(\bar{z})^5=(2+3i)^5+(2-3i)^5$

$=2({}^5C_0 2^5+{}^5C_2 2^3(3i)^2+{}^5C_4 2^1(3i)^4)$

$=2(32+10\times8(-9)+5\times2\times81)=244$

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